

Answers: Two Surveys from 1987

Shere Hite Survey

- a) **Population:** all American women in relationships with men. This is the group Hite drew conclusions about when she argued that dissatisfaction among American women was widespread.
- b) **Sample:** the approximately 4,500 women who returned completed questionnaires. **Sample size:** $n = 4,500$. The 100,000 questionnaires Hite distributed are the number sent out, not the sample.
- c) **No.** This is a voluntary response sample: the women decided for themselves whether to answer. Women with strong feelings — especially strong negative feelings — are far more likely to take the trouble to fill out and return a long questionnaire, so the respondents are not a cross-section of American women. The questionnaires were also distributed largely through women's organizations rather than to a random selection of women. The results describe the women who chose to answer, not American women generally.
- d) $4,500/100,000 = 4.5\%$. About 95.5% of the women who received a questionnaire never returned one, which leaves enormous room for the non-respondents to differ from the respondents.

Washington Post–ABC News Poll

- a) **Population:** all adults in the United States.
- b) **Sample:** the 1,505 adults interviewed nationwide by telephone. **Sample size:** $n = 1,505$.
- c) The difference comes from **how each sample was selected**, not from how large it was. Hite's 4,500 respondents were self-selected out of 100,000 questionnaires, a 4.5% response rate, and the women who chose to respond were disproportionately those unhappy enough to want to say so. That bias does not shrink as the sample gets bigger — a larger voluntary response sample is just a larger biased sample. The Washington Post–ABC News poll drew a random sample of adults nationwide, which is why it can report a margin of sampling error of about ± 3 percentage points; Hite's survey cannot honestly report one at all.

Two smaller differences are also worth noting. The surveys studied different populations: Hite surveyed women only, while the poll interviewed adults of both sexes. They also used different methods — an anonymous written questionnaire versus a telephone interview with a live person — and people are generally less willing to admit relationship problems to an interviewer than on paper, so the poll's very positive results may themselves be somewhat overstated. Even so, selection bias in Hite's sample is the main explanation for the gap.